Evaluation Report — Multi-Agent RAG Framework

3 November 2026

Summary

Both chunking strategies scored perfectly on retrieval: hit@5 and MRR were 1.000, answer accuracy 92.3%, faithfulness 100%, and 5/5 refusals with zero permission leaks. Both wrong answers came from an over-strict no-context threshold, not from retrieval. Recursive chunking is recommended. It tied on accuracy, had a lower median latency, and keeps document structure intact. The main weakness is speed: median end-to-end time is 7.5–8.3 s in evaluation and 13–14 s in the chat UI, driven by up to three Gemini calls per turn.

The build is a four-agent RAG system (Orchestrator, Ingestion, Retrieval, Synthesis) over Qdrant, Gemini and a cross-encoder re-ranker, with a React chat UI. Two chunking strategies, fixed-size and recursive, were indexed into separate Qdrant collections and compared on retrieval accuracy, faithfulness and latency.

Setup

Everything ran on a Heropa Windows Server 2022 VM, CPU only.

Component	Choice
Corpus	9 documents for a fictional company: 4 DOCX, 2 PDF, 3 Markdown (leave, remote work, travel, expenses, IT security, VPN, onboarding, salary bands, incident postmortem)
Chunk metadata	source, author, date, access_tags (all_employees, engineering, hr_only), chunk_index
Test set	26 questions: 21 answerable, 2 permission tests, 3 unanswerable
Embeddings	gemini-embedding-001
LLM	gemini-3.8-flash
Vector DB	Qdrant (local mode), one collection per chunking strategy
Retrieval	dense + BM25, merged with reciprocal rank fusion, re-ranked by cross-encoder ms-marco-MiniLM-L-6-v2, top 5; permission filter applied inside both searches
Synthesis	cited answer, then a per-sentence verification pass that removes unsupported sentences
Orchestrator	routes each turn to ingest, question or clarify; rewrites follow-ups into standalone queries

Configurations compared

The two strategies differ only in how documents are split. Everything else in the pipeline is identical, and each strategy has its own Qdrant collection.

Config	Collection	Splitter	Chunk size / overlap (chars)	Chunks indexed
A: Fixed-size	docs_fixed	CharacterTextSplitter, splits on spaces	500 / 50	11
B: Recursive	docs_recursive	RecursiveCharacterTextSplitter, paragraph → line → sentence → word	800 / 100	9 (one per document)

No document exceeds 800 characters, so the recursive splitter kept each one whole. The fixed-size splitter split two documents, hr_leave_policy.docx and vpn_setup_guide.md, into two chunks each. In practice the comparison is whole documents versus fixed 500-character windows.

Quantitative results

Both strategies scored identically on accuracy: perfect retrieval, 92.3% answer accuracy, 100% faithfulness and zero permission leaks. They differ only in latency. These scores come from the evaluation script over all 26 questions, top 5 chunks (run 2026-11-03 08:13).

Metric	A: Fixed-size	B: Recursive
Hit@5 (21 answerable)	100%	100%
Recall@5	1.000	1.000
MRR	1.000	1.000
Answer accuracy (LLM judge, 26 questions)	92.3% (24/26)	92.3% (24/26)
Faithfulness	100%	100%
Refusal accuracy (3 unanswerable + 2 permission)	100% (5/5)	100% (5/5)
Permission leaks	0	0
Retrieval latency p50 / p95 (ms)	1,087 / 1,218	1,074 / 1,159
End-to-end latency p50 / p95 (ms)	8,267 / 11,073	7,544 / 13,314

The evaluation calls the Retrieval and Synthesis agents directly and skips the orchestrator's routing call. That is why its end-to-end times are about 5 s lower than in the chat UI.

An earlier retrieval-only run (2026-11-03 08:11) gave the same retrieval scores for both strategies: hit@5, recall@5 and MRR all 1.000, with zero leaks. Its retrieval p50 / p95 was 1,011 / 1,097 ms for fixed-size and 1,001 / 1,074 ms for recursive. Retrieval quality and speed were stable across both runs.

Effect of the re-ranker

Turning off the cross-encoder halved retrieval time and cost no accuracy on this corpus. Both runs used the recursive collection (run 2026-11-03 08:25).

Metric	With re-rank	Without re-rank
Hit@5 / Recall@5 / MRR	1.000 / 1.000 / 1.000	1.000 / 1.000 / 1.000
Answer accuracy	92.3% (24/26)	100% (26/26)
Faithfulness	100%	100%

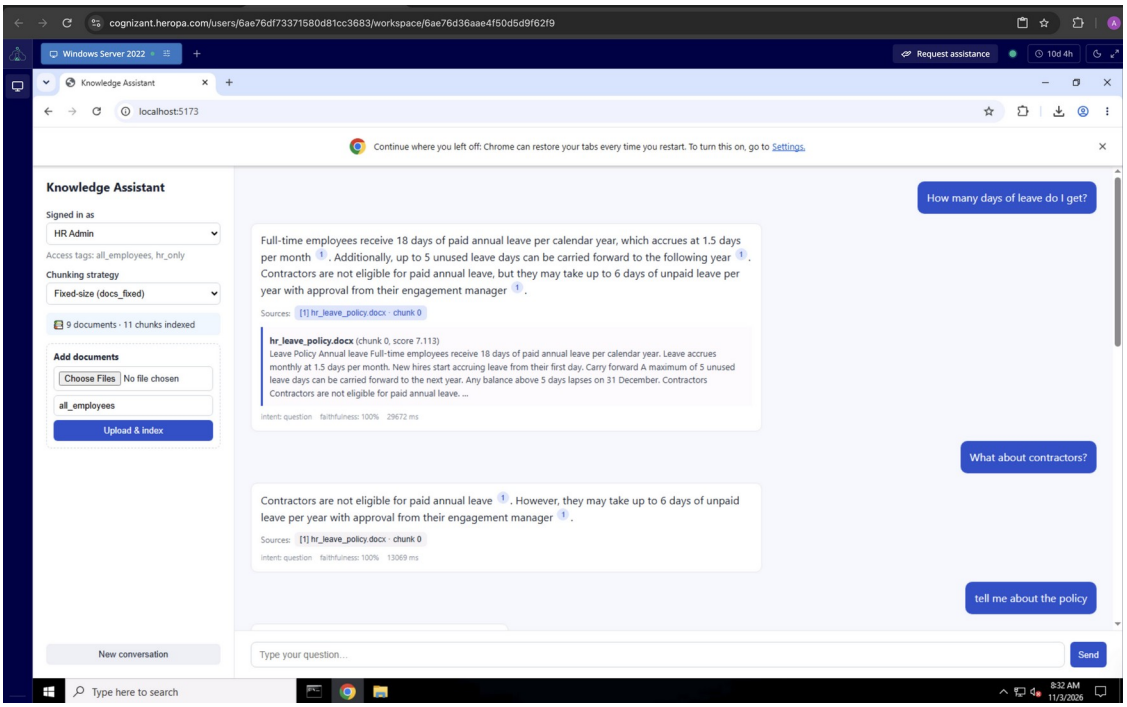
Metric	With re-rank	Without re-rank
Refusal accuracy	100% (5/5)	100% (5/5)
Permission leaks	0	0
Retrieval latency p50 / p95 (ms)	1,101 / 1,315	545 / 585
End-to-end latency p50 / p95 (ms)	7,747 / 11,146	7,997 / 10,344

Hybrid search plus RRF already ranked the right source first every time, so the re-ranker had nothing to fix. Its cost is about 0.55 s per query on CPU. The 2-question accuracy gap is not noise. q14 and q15 failed with the re-ranker and passed without it, because the no-context gate only applies when re-rank scores exist. Without the re-ranker, refusals rested entirely on the permission filter and the answer prompt. They still held at 5/5.

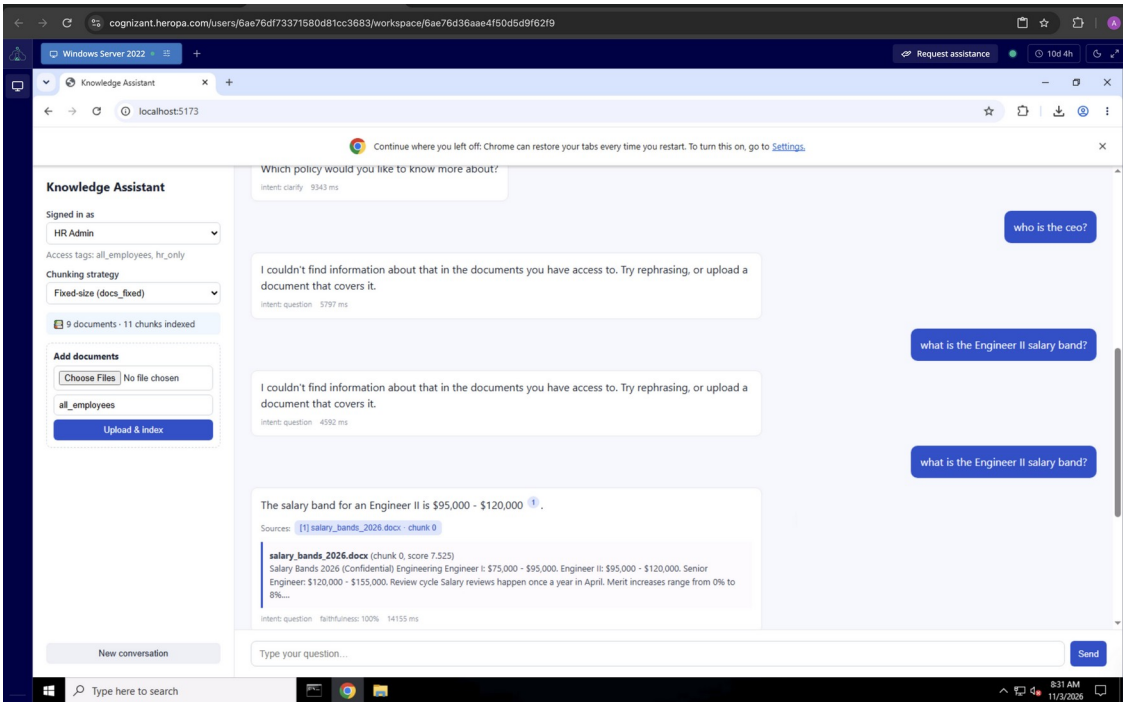
Functional test results

All six tests passed in the React chat UI, run against the fixed-size collection.

Test	Question	Role	Result	Latency (ms)
Cited answer	How many days of leave do I get?	HR Admin	18 days a year, cited to hr_leave_policy.docx chunk 0 (re-rank score 7.113); faithfulness 100%	29,672 (first query of the session)
Follow-up	What about contractors?	HR Admin	Resolved to contractors' leave: no paid leave, up to 6 unpaid days with approval; cited; faithfulness 100%	13,069
Clarification	tell me about the policy	HR Admin	Asked: "Which policy would you like to know more about?" (intent: clarify)	9,343
No context	who is the ceo?	HR Admin	Replied that the documents don't cover it; no citations	5,797
Permission denied	what is the Engineer II salary band?	Employee	Refused; the hr_only document was never retrieved	4,592
Permission granted	what is the Engineer II salary band?	HR Admin	$95,000 – $120,000, cited to salary_bands_2026.docx chunk 0 (score 7.525); faithfulness 100%	14,155



Cited answer with source snippet, and a follow-up question resolved from conversation state



Clarifying question, no-context reply, and the same salary question refused as Employee then answered as HR Admin

Analysis

Retrieval accuracy. Both strategies retrieved the right source first for all 21 answerable questions (hit@5 and MRR both 1.000), so on this corpus chunking made no difference to accuracy. The corpus is small: 9 short documents give 11 fixed-size chunks, and the recursive splitter kept all 9 documents whole. Where fixed-size chunking split a document, both halves were retrieved together, so nothing was lost. The first query of each run took 13.7 s to retrieve while the cross-encoder loaded; every later query took about 1 s. The cross-encoder separated the right chunk clearly: top chunks scored 7.1–7.5 against 2.8 for the next best.

Faithfulness. Faithfulness was 100% for both strategies: every sentence traced back to a retrieved chunk. Answer accuracy was 92.3%, with 2 of 26 judged wrong in each configuration. Both configurations missed the same two questions, q14 and q15, both answered by it_security_runbook.md. The runbook ranked first both times, yet the system replied that it couldn't find the answer. The cause is the no-context gate, not search (see Failure examples). The weak point is over-answering. Asked only how many leave days you get, the system also explained carry-forward and contractor rules. That text was grounded but not asked for.

Latency. Latency tracks the number of Gemini calls each path makes:

Path	Gemini calls	Observed (s)
No context / permission denied	1 (router), then retrieval stops early	4.6 – 5.8
Clarification	1 (router)	9.3
Answered question	3 (router, answer, verifier)	13.1 – 14.2
First question of the session	3, plus loading the cross-encoder	29.7

The answer and verifier calls add roughly 8–9 s per answered question. Retrieval takes only about 1.1 s (p50) under both strategies, so the Gemini round trips are the bottleneck, not the vector search. Recursive chunking had a lower median (7.5 s vs 8.3 s) but a higher p95 (13.3 s vs 11.1 s). With retrieval times almost identical, that spread most likely reflects Gemini response-time variance, not the chunking.

Failure examples

Both failures were refusals of questions the corpus does answer, and both failed in both chunking configurations.

ID	Question	Expected	Got	Likely cause	Fix
q14	What is the minimum password length?	14 characters	"I couldn't find information about that…"	it_security_runbook.md ranked 1st, but its cross-encoder score fell below the −1.0 no-context threshold	Calibrate MIN_RERANK_SCORE on the test set (e.g. −3.0)
q15	What should I do if I lose my laptop?	Call the IT helpdesk on ext. 4357; IT remotely locks and wipes the device	"I couldn't find information about that…"	Same: correct source ranked 1st, answer blocked by the threshold	Same

The final run (08:25) confirms the cause. With the re-ranker and its threshold off, the same top-ranked runbook chunk produced correct, cited answers: “Passwords must be at least 14 characters long [1]” and “call the IT helpdesk immediately on extension 4357 [1]. IT will then remotely lock and wipe the device [1].” The runbook packs three topics into one short document: passwords, phishing and lost laptops. That probably dilutes how relevant the cross-encoder judges it to any one question.

Limitations

- **Small corpus.** Nine short documents give too few chunks to separate the two chunking strategies clearly. A larger corpus with long, multi-section documents would test chunking properly.
- **Latency.** Answered questions take 13–14 s, which is slow for a chat UI.
- **Verbose answers.** Answers can include grounded but unrequested detail.
- **LLM-judged metrics.** Answer accuracy and faithfulness are scored by Gemini itself, so they share the model’s blind spots.
- **In-memory state.** Conversation state is lost when the server restarts.
- **Keyword index rebuilt per role.** BM25 is rebuilt in memory for each role. That’s fine for 11 chunks but won’t scale; Qdrant’s native sparse vectors would.

Conclusion and next steps

The pipeline meets every requirement of the case study. It retrieves perfectly, gives 100% faithful cited answers with 92–100% accuracy, refused all 5 questions it should have refused and leaked nothing across permissions. The two chunking strategies tied on every accuracy metric. Recursive chunking is recommended: it keeps sections and sentences intact, which will matter as documents grow. The re-ranker added about 0.55 s per query without improving retrieval here, and its no-context threshold was set too strictly, causing both wrong answers. Keep it, but calibrate the threshold.

Next steps, in priority order:

1. Calibrate the no-context threshold (MIN_RERANK_SCORE, currently −1.0) against the test set, so that q14 and q15 pass and the 5 refusals still hold.
2. Cut latency by merging the router into a single cheaper call, or by skipping it for clear questions.
3. Tighten the answer prompt to answer only what was asked.
4. Grow the corpus with longer documents and re-run the chunking comparison.
5. Persist conversation state, for example in Redis or Postgres.